FAST – National University of Computer and Emerging Sciences

Theory of Automata

Assignment # 2

Deadline: 31 October 2025 Instructions:

1. Attempt all questions.
2. Questions must be attempted on paper (A4 sheets or loose sheets) 3. Submit the assignment on time. Only physical submissions are accepted.

Design Context Free Grammar and PDA for given Context Free Language.

I. To design a context free grammar which can produce balance parenthesis consider in alphabet set we have { (,)}

   S   S S | "(" S ")" | ε

II. Design a CFG and PDA for even and odd palindrom
   a. Even palindrome 001 100
      S   0 S 0 | 1 S 1 | ε
   b. Odd palindrome 00100

      S   0 S 0 | 1 S 1 | 0 | 1

III. Design a CFG and PDA for language $\{01(1100)^n\ 110\ (10)^n \mid n>=0\}$

   S   01 D
   D   1100 D 10|110

IV. Design a CFG and PDA for language $\{ o^n\ 1^m \mid n!= m\}$ means {n>m,n<m}

   S   S1 | S2
   S1   0S1 1 | 0S1 | 0
   S2   0S2 1 | S2 1 | 1

V. Design a CFG and PDA for language $\{ (0^n\ 1^n\ 2^m\ 3^m \mid n>=1, m>=1)$ or $(0^n\ 1^m\ 2^m\ 3^n \mid n>=1, m>=1) \}$

S A | B
A X Y
X 0 1 | 0 X 1 (produces 0^n1^n, n≥1)
Y 2 3 | 2 Y 3 (produces 2^m3^m, m≥1)
B P Q
P 0 3 | 0 P 3 (produces 0^n3^n, n≥1)
Q 1 2 | 1 Q 2 (produces 1^m2^m, m≥1)

VI. Design a CFG and PDA for language { (0^i 1^j 2^k | k <= i,OR k <= j)

S X| Y
(Case k ≤ i)
X BCD
B 0B | ε
C D | 0 C 2
D 1 D | ε
(Case k ≤ j) — symmetric: swap roles of 0 and 1 in the above
Y Z LM
Z 0 Z | ε
L M | 1 L 2
M 1 M | ε